1

Recurrence: Solve $T(n) = 5T(n/7) + 3n^2$ w/ substitution

$T(n) = 5T(n/7) + 3n^2$

$T(n/7) = 5T(n/49) + 3(n/7)^2$

substitute

$T(n) = 5[5T(n/49) + 3(n/7)^2] + 3n^2$

$= 5^2 T(n/7^2) + 3(5/49)n^2 + 3n^2$

2nd sub

$T(n) = 5^3 T(n/7^3) + 3(5/49)^2 n^2 + 3(5/49)n^2 + 3n^2$

$n/7^k = 1$

$n = 7^k$

$k = \log_7(n)$

Pattern:

$T(n) = 5^k \cdot T(n/7^k) + 3n^2 \cdot \sum_{i=0}^{k-1} (5/49)^i$

sub $k = \log_7(n)$

$T(n) = \underbrace{5^{(\log_7(n))}}_{= n^{(\log_7 5)}} \cdot \overbrace{T(1)}^{\text{constant}} + \underbrace{3n^2 \cdot \sum_{i=0}^{(\log_7 n - 1)} (5/49)^i}_{3n^2 \cdot \Sigma = \Theta(n^2)}$

$= c \cdot n^{(\log_7 5)} + \Theta n^2$

$T(n) = \Theta n^2$

Masters: $a = 5$, $b = 7$, $f(n) = 3n^2$

Case 3: $3n^2$ grows faster than $n^{(\log_7 5)} \approx n^{(0.83)}$

$a \cdot f(n/b) \leq c \cdot f(n)$

$5 \cdot 3(n/7)^2 \leq c \cdot 3n^2$ div $3n^2$

$15n^2/49 < c \cdot 3n^2 \rightarrow 15/49 < c$

$c = 0.5$ ✓ $15/49 \approx 0.31 < 0.5 < 1$

$T(n) = \Theta(fn) = \Theta(n^2)$.

2

a) $T(n) = 5T(n/7) + n^7$

$a = 5,\ b = 7,\ n^{(\log_7 5)} = n^{(0.83)}$

$f(n) = n^7$ grows faster than $n^{(0.83)} \rightarrow$ case 3

$5(n/7)^7 = 5n^7/7^7 \leq cn^7 \quad (c = 5/7^7 < 1)$

$T(n) = \Theta(n^7)$

b) $T(n) = 3T(n/3) + 7n$

$a = 3,\ b = 3,\ n^{(\log_3 3)} = n^1 = n$

$f(n) = 7n = \Theta(n) \rightarrow$ case 2

$T(n) = \Theta(n \log n)$

c) $T(n) = 15T(n/2) + 100$

$a = 15\ T\ 1 = 2,\ n^{\log_2 15} = n^{3.91}$

$f(n) = 100 \Rightarrow$ case 1

$T(n) = \Theta(n^{\log_2 15})$

d) $T(n) = 9T(n/3) + n^2$

$a = 9,\ b = 3,\ n^{\log_3 9} = n^2$

$f(n) = n^2 \Rightarrow$ case 2

$T(n) = \Theta(n^2 \log n)$

e) $T(n) = 6T(n/5) + n^3$

$a = 6,\ b = 3,\ n^{\log_3 6} = n^{1.11}$

$f(n) = n^3$ grows faster than $n^{1.11} \Rightarrow$ case 3

$6n^3/125 \leq cn^3 \quad (c = 6/125 < 1)$

$T(n) = \Theta(n^3)$

3) Radix Sort

BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,LA_GUA,

TOL,META,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR

First pass: (move LA_GUA to the end and all others are padded with null)

BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,

TOL,META,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR, LA_GUA

2nd pass: (move META)

BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,

TOL,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR, META,LA_GUA\

Pass 3 (sort all of them into buckets of the 3rd index)

LA_GUA, GUA, GUA, AMA, NSA, SUC, VIC, MAG, QUI, HUI, ATL, VAL, CAL, BOL, TOL, ANT, SAN, SAN, CHO, CAQ, COR, RIS, CES, CAS, PUT, META, VAU, CAU, BOY

4th pass (now sort based off of second index)

LA_GUA, MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU, CES, META, CHO, VIC, RIS, AMA, ANT, BOL, TOL, COR, BOY, NSA, ATL, GUA, GUA, SUC, QUI, HUI, PUT

Final pass (sort on index 0)

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, LA_GUA, MAG, META, NSA, PUT, QUI, RIS, SAN, SAN, SUC, VAL, VAU, VIC

4) Double Hashing

7) time and space

4

5)

Time complexity is O(n*k) where n is the length of the array and k is the length of the longest input. We must go through all n strings in the input array and we the outer loop loops at most K times for the largest string length.

The space complexity is O(n) where n is the size of the input array because all of the string are stored into the input array.

6)

The time complexity is O(n^3) where n is the number of words because there are three nested loops each running n times.

The Space complexity is O(n^3) because, in a scenario where every possible combination of three words forms a palindrome, you must store all unique triplets in your result list.